



INDUSTRY

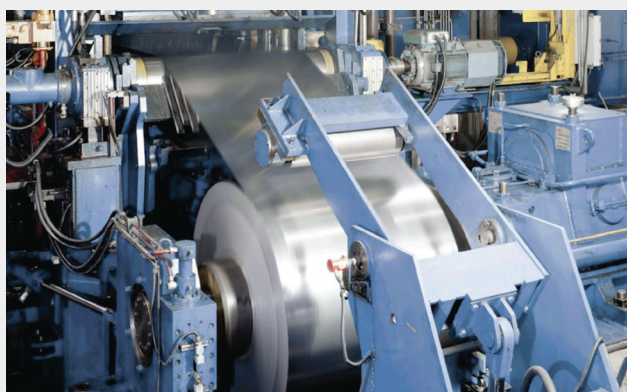
STEEL AND METALS

APPLICATION

COLD ROLLING MILL

COST SAVINGS: \$28,288

A major steel producer was dissatisfied with the performance of their Four-Row Tapered Roller Bearings installed in their cold rolling mill. Steel shavings from the production process were entering the bearings resulting in a detrimental effect on bearing reliability. NSK performed a detailed on-site analysis of the bearings along with inspecting a failed bearing and lubrication analysis. After thorough review, a recommendation was made to adopt NSK Tough Steel technology which offers an excellent resistance to damage caused by the ingress of contamination.



Cold Rolling of Steel

KEY FACTS

- › Cold rolling mill
- › Damage of the bearings due to ingress of steel shavings from the production process
- › NSK solution: Bearings of Water Tough (WTF) steel
- › More than 3X longer bearing life - reducing bearing consumption, downtime and maintenance costs

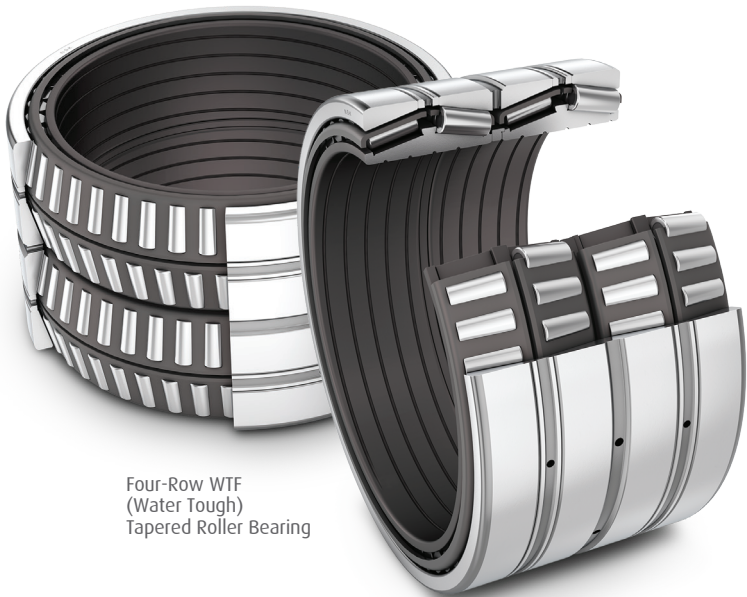
VALUE PROPOSALS

- › Analysis of the bearings, grease, structure and roll design
- › NSK WTF steel bearings - designed for extreme operating conditions including solid and liquid contamination - were recommended
- › Technical support including on-site engineering consultancy and lab-based bearing analysis



PRODUCT FEATURES

- › Advanced WTF (Water-Tough) material composition containing appropriate levels of chrome and molybdenum for increased hardness
- › Innovative and patented heat treatment technology to optimize retained austenite and formation of finer carbide and carbonitride particles
- › Significantly outperforms standard bearing steel in seizure resistance, rate of wear and service life
- › Available for four-row cylindrical and four-row taper roller bearings
- › Bearing life is 3 times longer than that of conventional bearing
- › Grain boundaries have been strengthened to help prevent the propagation of cracks



COST-SAVING BREAKDOWN

BEFORE	COST	NSK SOLUTION	COST
 Bearing life of the previous bearings: 4 months, Lost production: \$1,071/hr downtime X number of replacements X 4 unexpected downtimes per year	\$47,147	Bearing life of the new bearings: 12 months Lost production: \$1,071/hr downtime X number of replacements X 1 unexpected downtime per year	\$18,859
TOTAL COSTS			
	\$47,147		\$18,859

YOUR PARTNER FOR MACHINE OPTIMIZATION

Our AIP Added Value Program is based around a simple proposition: ‘improvement pays’. By working with you throughout the AIP Value Cycle, we will help you achieve improvements in machine reliability, productivity and performance, all of which carry a tangible and measurable cost benefit – and we have the tools to prove it! That’s what we mean by **improvement pays**.

